

Quick Check

Lesson 2: Defining Sine, Cosine, and Tangent — Naming the Ratios That Unlock the Mast

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.4: Trigonometry 1 | Lesson 2

A student stands 40 m from the base of a mast and measures the angle from the ground up to the top as 50° . The mast's height is the side opposite the 50° angle, and the 40 m is the side adjacent to it. Which ratio links the angle to the height?

A $\sin 50^\circ = \text{opposite} \div \text{hypotenuse}$

B $\cos 50^\circ = \text{adjacent} \div \text{hypotenuse}$

C $\tan 50^\circ = \text{opposite} \div \text{adjacent}$

D $\tan 50^\circ = \text{adjacent} \div \text{opposite}$

In a right-angled triangle PQR, the right angle is at Q and PR is the longest side. Looking from angle P, a learner labels side QR as 'opposite'. She then switches to look from angle R. What is side QR called now?

- A Still opposite, because the side does not move
- B Adjacent, because it now lies next to the reference angle R
- C It has no label, because it is only labelled from angle P
- D Hypotenuse, because it is a side of the triangle

Two learners draw a right-angled triangle with an angle of about 35° . They measure the opposite side as 5.8 cm and the hypotenuse as 10.0 cm, so their $\sin 35^\circ$ is 0.58. The mathematical table gives $\sin 35^\circ = 0.574$. What best explains the small difference?

- A** Small errors in drawing and measuring the sides; the table value is the reliable one
- B** The table is wrong because it does not match their triangle
- C** Their triangle is too small, so the sine ratio changes with size
- D** They should have used adjacent $\div$ hypotenuse for sine

A 10 m ladder leans against a wall and makes an angle of 60° with the ground. The ladder is the hypotenuse, and the height it reaches up the wall is the side opposite the 60° angle. Which ratio should be used to find that height?

A $\tan 60^\circ = \text{adjacent} \div \text{opposite}$

B $\tan 60^\circ = \text{opposite} \div \text{adjacent}$

C $\sin 60^\circ = \text{opposite} \div \text{hypotenuse}$

D $\cos 60^\circ = \text{adjacent} \div \text{hypotenuse}$

A student stands 40 m from the base of a mast and measures the angle of elevation to the top as 50° . From tables, $\tan 50^\circ \approx 1.192$. What is the height of the mast, to 1 decimal place?

A 25.7 m

B 30.6 m

C 33.6 m

D 47.7 m

A surveyor in Kisumu stands 25 m from the base of a tree and measures the angle of elevation to the top as 38° . Using $\tan 38^\circ \approx 0.781$, what is the height of the tree, to 1 decimal place?

A 15.4 m

B 40.6 m

C 32.0 m

D 19.5 m

You know that $\sin \theta = 0.643$ for an acute angle θ and want to find θ using mathematical tables. What should you do?

- A** Divide 0.643 by 90 to get the angle
- B** Find 0.643 in the sine table and read the angle that goes with it
- C** Find 0.643 in the cosine table and read the angle that goes with it
- D** Multiply 0.643 by 100 to get the angle in degrees

You are adding ratio formulas to your right-angled triangle model, which has sides labelled opposite (O), adjacent (A) and hypotenuse (H) relative to angle θ . Which formula belongs next to the adjacent side and the hypotenuse?

A $\tan \theta = A \div H$

B $\cos \theta = O \div H$

C $\cos \theta = A \div H$

D $\sin \theta = A \div H$

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).